Assignment 1

CSE260: Digital Logic Design

Department of Computer Science and Engineering

Semester : Spring 26 **Marks: 5**

<u>Graded (5 marks - 1 mark each)</u>

1. Convert the decimal number $(9384)_{10}$ to its **Binary**, **BCD**, and **Excess-4** representations and prove that the binary, BCD, and Excess-4 representations are not equivalent. You must show the conversion or required calculations for every step.

2. Convert the hexadecimal number $(7A3.C9)_{16}$ to its base-5 equivalent. Must show all the steps during conversion.

3. Evaluate the following **without converting to decimal**:
   a. $(3A4)_{16} + (257)_8 - (110101)_2$
   Express the final answer in **hexadecimal**.

   b. Perform $(321654325)_7$ / $(5433)_7$
   Find the quotient and remainder.

4. Add -71 and -63 using 8 bits in 2's complement. Determine if overflow occurs and justify your answer. Mention the valid range for 8 bits and verify whether the result falls within range or not.

5. You are planning an event and need to buy the following items:
   - A banner costs $(101101)_2$ dollars
   - Lighting costs $(57)_8$ dollars
   - Sound system rental costs $(2D)_{16}$ dollars
   - 6 food boxes, each costing $(111)_2$ dollars

   You have $(1110101)_2$ dollars in your wallet. Calculate the **total cost** Determine **how much extra money** you need. Express the final answer **in decimal**.